

CS1231S: Discrete Structures
Tutorial #5: Relations & Partial Orders
 (Week 7: 28 September – 2 October 2026)
Answers

II. Tutorial Questions

1. Let S be the set of all strings over the alphabet $\mathcal{A} = \{s, u\}$, i.e. an element of S is a sequence of characters, each of which is either s or u . Examples of elements of S are: ε (the empty string), $s, u, sus, usssuu$, and $susussssu$.

Define a relation R on S by the following: $\forall a, b \in S (aRb \Leftrightarrow \text{len}(a) \leq \text{len}(b))$ where $\text{len}(x)$ denotes the length of x , i.e. the number of characters in x .

Is R a partial order? Prove or disprove it.

Answer:

R is not antisymmetric, hence it is not a partial order. Counterexample: $s R u$ and $u R s$ but $s \neq u$.

Let R be a relation on a set A .

R is **antisymmetric** iff $\forall x, y \in A (xRy \wedge yRx \Rightarrow x = y)$.

2. [AY2022/23 Semester 1 Midterm Test]

Let $A = \{2, 3, 5, 7, 21, 30, 84, 99\}$ and let $\preceq$ be a partial order on the set A defined by the “divides” relation, that is, $x \preceq y \Leftrightarrow x|y$. Which of the following statements are true?

- (i) The partial order has a linearization $\preceq^*$ such that $21 \preceq^* 7$.
- (ii) The partial order has a linearization $\preceq^*$ such that $3 \preceq^* 2$.
- (iii) The partial order has a linearization $\preceq^*$ such that $21 \preceq^* 5 \preceq^* 84$.
- (iv) The partial order has a linearization $\preceq^*$ such that $99 \preceq^* 84 \preceq^* 30$.

Answer: Only (ii), (iii) and (iv) are true.

2 and 3 are noncomparable, 5 and 21 are noncomparable, 30, 84 and 99 are noncomparable, so they can appear in any relative order in the linearization.

3. [AY2023/24 semester 1 Midterm Test]

Let $A = \{11, 12, 13, 14, 15, 16\}$. For each $x \in A$, define $F_x = \{k \in \mathbb{Z}^+ : k|x\}$, where $|$ is the “divides” relation. Define also a partial order $\preceq$ on A by setting for all $x, z \in A$:

$$x \preceq z \Leftrightarrow (F_x = F_z) \vee (|F_x| < |F_z|).$$

What are the minimal, smallest, maximal, and largest elements of A with respect to $\preceq$?

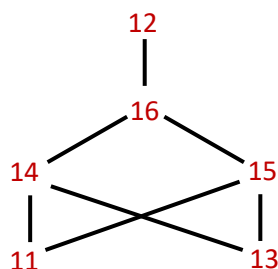
Answers:

Minimal: 11, 13.

Smallest: None.

Maximal: 12.

Largest: 12.



Let a set A be partially ordered with respect to a relation $\preceq$ and let $c \in A$.

- c is a **maximal element** of A iff $\forall x \in A (c \preceq x \Rightarrow c = x)$.
- c is a **minimal element** of A iff $\forall x \in A (x \preceq c \Rightarrow c = x)$.
- c is the **largest element** of A iff $\forall x \in A (x \preceq c)$.
- c is the **smallest element** of A iff $\forall x \in A (c \preceq x)$.

4. Given the partial order $\leq$ on A in question 3 above, one of the linearizations $\leq^*$ of $\leq$ is shown below:

$$11 \leq^* 13 \leq^* 14 \leq^* 15 \leq^* 16 \leq^* 12$$

Write out all the other possible linearizations $\leq^*$.

Answers: There are 4 possible linearizations; the other 3 are:

$$11 \leq^* 13 \leq^* 15 \leq^* 14 \leq^* 16 \leq^* 12$$

$$13 \leq^* 11 \leq^* 14 \leq^* 15 \leq^* 16 \leq^* 12$$

$$13 \leq^* 11 \leq^* 15 \leq^* 14 \leq^* 16 \leq^* 12$$

5. Let $B = \{0,1\}$ and define the binary relation R on $B \times B$ as follows:

$$\forall (a,b), (c,d) \in B \times B \left((a,b) R (c,d) \Leftrightarrow (a \leq c) \wedge (b \leq d) \right).$$

- Prove that R is a partial order.
- Draw the Hasse diagram for R .
- Find the maximal, largest, minimal and smallest elements.
- Is $(B \times B, R)$ well-ordered?

Answers:

(a) Proof:

- (Reflexivity) Take any $(a,b) \in B \times B$,

1.1. $a \leq a$ and $b \leq b$.

1.2. So $(a,b) R (a,b)$ by the definition of R .

1.3. Hence R is reflexive.

- (Antisymmetry) Take any $(a,b), (c,d) \in B \times B$,

2.1. Suppose $(a,b) R (c,d)$ and $(c,d) R (a,b)$.

2.2. Then $a \leq c, b \leq d, c \leq a$ and $d \leq b$ by the definition of R .

2.3. Then $a = c$ and $b = d$ by the antisymmetry of $\leq$.

2.4. So $(a,b) = (c,d)$ by equality of ordered pairs.

2.5. Hence R is antisymmetric.

- (Transitivity) Take any $(a,b), (c,d), (e,f) \in B \times B$,

3.1. Suppose $(a,b) R (c,d)$ and $(c,d) R (e,f)$.

3.2. Then $a \leq c, b \leq d, c \leq e$ and $d \leq f$ by the definition of R .

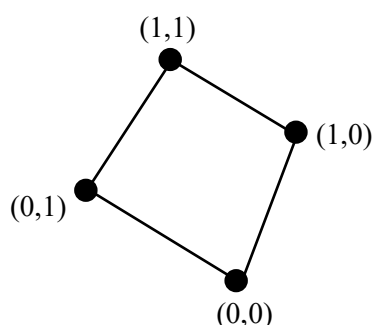
3.3. Then $a \leq e$ and $b \leq f$ by the transitivity of $\leq$.

3.4. So $(a,b) R (e,f)$ by the definition of R .

3.5. Hence R is transitive.

- Therefore R on $B \times B$ is a partial order.

(b)



(c) Maximal and largest: $(1,1)$;
Minimal and smallest: $(0,0)$.

(d) No. It is not even a total order, because $(0,1)$ and $(1,0)$ are not comparable.

R is a total order on a set A iff
 R is a partial order on A and $\forall x, y \in A (xRy \vee yRx)$.

6. Let R be a binary relation on a non-empty set A . Let $x, y \in A$. Define a relation S on A by

$$x S y \Leftrightarrow (x = y) \vee (x R y) \text{ for all } x, y \in A.$$

Show that:

- (a) S is reflexive;
- (b) $R \subseteq S$; and
- (c) if S' is another reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$.

What is this relation S called? (Hint: Refer to Transitive Closure in Lecture 6).

Answers:

- (a) 1. Let $x \in A$.
2. $x = x$, so $x S x$ by the definition of S .
3. Therefore, S is reflexive.
- (b) 1. Suppose $(x, y) \in R$, that is, $x R y$.
2. So $x S y$ by the definition of S .
3. So $(x, y) \in S$.
4. Therefore, $R \subseteq S$ by the definition of $\subseteq$.
- (c) 1. Suppose $(x, y) \in S$.
2. Then $x S y$, which means $x = y \vee x R y$ by the definition of S .
3. Case 1: $x = y$
3.1. Then $x S' y$ since S' is reflexive.
3.2. So $(x, y) \in S'$.
4. Case 2: $x R y$
4.1. Then $(x, y) \in R \subseteq S'$.
4.2. Then $(x, y) \in S'$.
5. In all cases, $(x, y) \in S'$.
6. Therefore, $S \subseteq S'$.

Let R be a relation on a set A .
 R is **reflexive** iff $\forall x \in A (x R x)$.

The relation S is called the **reflexive closure** of R . It is the smallest relation on A that is reflexive and contains R as a subset.

7. Let R be a binary relation on a set A .
We have defined antisymmetry in class: R is **antisymmetric** iff

$$\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y).$$

We define asymmetry here. R is **asymmetric** iff

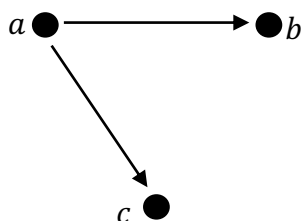
$$\forall x, y \in A (x R y \Rightarrow y \not R x).$$

- (a) Find a binary relation on A that is both asymmetric and antisymmetric.
- (b) Find a binary relation on A that is not asymmetric but antisymmetric.
- (c) Find a binary relation on A that is asymmetric but not antisymmetric.
- (d) Find a binary relation on A that is neither asymmetric nor antisymmetric.

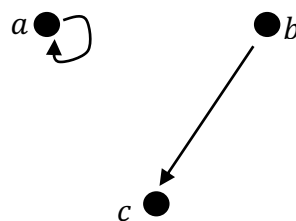
Answers:

Let $A = \{a, b, c\}$.

(a) $R = \{(a, b), (a, c)\}$.



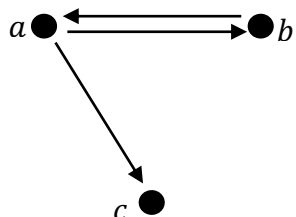
(b) $R = \{(a, a), (b, c)\}$



(c) No solution.

Every asymmetric relation is antisymmetric (see proof below).

(d) $R = \{(a, b), (b, a), (a, c)\}$



Proof: (Every asymmetric relation is antisymmetric.)

1. Take any binary relation R on a set A .
2. Suppose R is asymmetric.
 - 2.1. Then $\forall x, y \in A (x R y \Rightarrow y \not R x)$ by the definition of asymmetry.
 - 2.2. $\equiv \forall x, y \in A (x \not R y \vee y \not R x)$ by the implication law.
 - 2.3. $\Rightarrow \forall x, y \in A ((x \not R y \vee y \not R x) \vee x = y)$ by generalization.
 - 2.4. $\equiv \forall x, y \in A (\sim(x R y \wedge y R x) \vee x = y)$ by the De Morgan's law.
 - 2.5. $\equiv \forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$ by the implication law.
3. Line 2.5 is the definition of antisymmetry, hence R is antisymmetric.

We see that asymmetry property forces the antisymmetry property to be vacuously true

8. **Definitions.** Consider a partial order $\leq$ on a set A and let $a, b \in A$.

- We say a, b are **comparable** iff $a \leq b$ or $b \leq a$.
- We say a, b are **compatible** iff there exists $c \in A$ such that $a \leq c$ and $b \leq c$.

Consider the “divides” relation on $A = \{1, 2, 4, 5, 10, 15, 20\}$. List out the pairs of distinct elements in A that are (a) comparable; (b) compatible. Use the notation $\{x, y\}$ to represent the pair of elements x and y .

Are two comparable elements always compatible? Are two compatible elements comparable?

Answers:

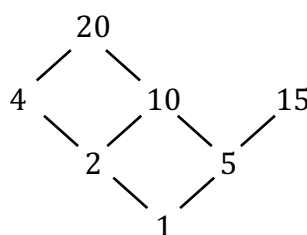
Hasse diagram

(a) Comparable:

$\{1, 2\}, \{1, 5\}, \{1, 4\}, \{1, 10\}, \{1, 15\}, \{1, 20\},$
 $\{2, 4\}, \{2, 10\}, \{2, 20\}, \{5, 10\}, \{5, 15\}, \{5, 20\},$
 $\{4, 20\}$ and $\{10, 20\}$.

Not comparable: $\{2, 5\}, \{2, 15\}, \{5, 4\}, \{4, 10\}, \{4, 15\}, \{10, 15\}$ and $\{15, 20\}$.

(b) Compatible:



$\{1,2\}, \{1,5\}, \{1,4\}, \{1,10\}, \{1,15\}, \{1,20\},$
 $\{2,4\}, \{2,5\}, \{2,10\}, \{2,20\}, \{5,4\}, \{5,10\}, \{5,15\}, \{5,20\},$
 $\{4,10\}, \{4,20\}$ and $\{10,20\}.$

Not compatible: $\{2,15\}, \{4,15\}, \{10,15\}$ and $\{15,20\}.$

(c) Yes, comparable elements are always compatible. If $\{a, b\}$ is comparable we can always chose $c = b$, such that c is greater than both elements. This is possible since partial order satisfies reflexivity.

(d) No. Counterexample: $\{2,5\},$

8. [AY2022/23 Semester 2 Mid-term Test]

Let $\leq$ be a partial order on a non-empty set A . A subset C of A is called a **chain** if and only if every pair of elements in C is comparable, that is, $\forall a, b \in C (a \leq b \vee b \leq a)$. A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain. The length of a chain is one less than the number of elements in it.

(a) Let $A = \{a, b, c, d\}$ and $(\mathcal{P}(A), \subseteq)$ be a poset on $\mathcal{P}(A)$, where $\mathcal{P}(A)$ denotes the power set of A . Write out two maximal chains in $(\mathcal{P}(A), \subseteq)$.

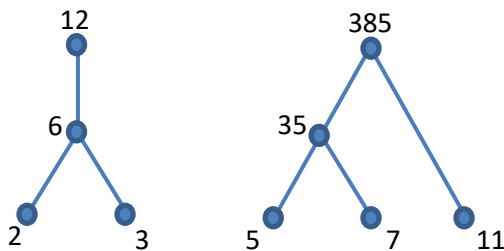
(b) Let $B = \{2,3,5,6,7,11,12,35,385\}$ and $(B, |)$ be a poset on B , where $|$ denotes the divides relation. Draw the Hasse diagram and write out two maximal chains of different lengths in $(B, |)$.

Answers:

(a) Many possible answers.

Two of them are $\{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}, \{a, b, c, d\}\}$ and $\{\emptyset, \{b\}, \{b, d\}, \{b, c, d\}, \{a, b, c, d\}\}.$

(b) Hasse diagram:



Maximal chains of different lengths:

$\{2,6,12\}$ and $\{11,385\},$
 or $\{3,6,12\}$ and $\{11,385\},$
 or $\{5,35,385\}$ and $\{11,385\},$
 or $\{7,35,385\}$ and $\{11,385\}.$